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The EU road refrigerated transport: current GHG footprint of transport refrigeration unit and projected impact of sustainable technologies

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ABSTRACT

The global road transport refrigeration fleet serving the food supply chain is estimated at approximately 5.7 million vehicles. This sector is experiencing continuous growth, driven by evolving consumer habits such as the rise in e-commerce and the increasing incidence of short-distance deliveries. Sectoral emissions were estimated at approximately 50 Mt CO_2eq in 2021, with the majority attributable to diesel-powered traction systems. Approximately 20 % of these emissions originate from the energy consumption of refrigeration units. At the same time, the entire road transport sector is under increasing pressure to meet stringent environmental targets. Moreover, working fluids used in vapor compression cycles are undergoing regulatory scrutiny, with a primary focus on reducing greenhouse gas (GHG) emissions, while growing attention is being paid to other potential environmental and human health impacts. This study investigates high technology-readiness-level (TRL) solutions aimed at reducing the overall environmental footprint of road transport refrigeration. The solutions considered include improved thermal insulation of refrigerated boxes, the adoption of natural refrigerants, advanced control strategies under partial load conditions, vehicle electrification, and the integration of renewable energy sources. The potential effects of implementing these technologies across the European refrigerated fleet are numerically assessed in terms of primary energy consumption, carbon-equivalent emissions, and pollutant emissions. Results indicate that the adoption of these technologies could yield reductions of up to 28 % in annual primary energy consumption, up to 72 % in CO_2-equivalent emissions, and over 90 % in emissions of other air pollutants.

1. Introduction

According to IIR (2021), approximately 70 % of food intended for human consumption requires refrigeration at some stage of its lifecycle. Expanding and strengthening the cold chain is therefore recognized as a key strategy to ensure adequate and safe nutrition for a growing global population. However, current cold chain infrastructure accounts for a significant share of global greenhouse gas emissions, estimated at 4 % including emissions from food loss and waste attributable to insufficient refrigeration (UNEP and FAO, 2022). IIR (2021) further estimates that the implementation of an "improved" cold chain, characterized by wider coverage and enhanced performance, could reduce these emissions by approximately 50 %. Moreover, the global expansion of cold chain infrastructure has the potential to prevent the loss of >475 million tons of food per year, sufficient for feeding approximately 950 million people annually (Baha et al., 2025).

Transport refrigeration represents a critical link of the cold chain, occurring at multiple stages throughout a product lifecycle and playing a pivotal role in preserving its quality and integrity. In 2021, IIR estimated in 3.4 million vehicles the global refrigerated fleet; a more recent estimation increased this number to 5.7 million of refrigerated road vehicles in service worldwide, including vans, trucks, semi-trailers and trailers (Baha et al., 2025), with projections of significant increase especially in emerging markets (+20 % in India and China over the current decade). The annual global emissions related just to transport refrigeration are estimated at approximately 50 Mt CO2eq, representing nearly 25 % of the total cold chain emissions (IIR, 2021).

The standard solution currently adopted in road refrigerated transport consists of an insulated box coupled with a cooling unit. The insulated box is built using sandwich panels, consisting of external layers of fibre-glass reinforced plastic containing insulating foam, locally reinforced with wood or metal to support mechanical loads. The transport refrigeration unit works according to a simple vapor compression cycle using synthetic refrigerant. The unit is powered by

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the vehicle engine in most Light Commercial Vehicles (LCV) or by a dedicated diesel engine in larger trucks.

To meet the general cold chain sustainability goals (UNEP and FAO, 2022) and specifically the call for neutral or positive environmental impact of the food chain, as required by the European Farm to Fork Strategy, it is relevant to analyse each component of the refrigeration equipment (insulated box and cooling unit) contributing to the emissions, as well considering also the vehicle prime engine and potential exploitation of renewable energy sources, to evaluate possible improvement strategies. The reduction of the thermal load is strictly linked to the insulation of the box, which is described by the overall heat transfer coefficient. This coefficient accounts for both distributed and localized heat transfer phenomena. In the present study, the definition provided by the ATP agreement (United Nations, 2024) is adopted, along with the reference values specified for Normal and Heavy Insulated boxes. Boundary conditions (outdoor temperature and humidity, solar radiation, indoor setpoint) and the overall mass of the insulated box affect the pulldown energy, since pulldown operations must be performed at empty box, before loading the products to be transported. On the other hand, boundary conditions, mission profile (overall distance, driving cycles, delivery mission structure, etc.) and transported goods thermal capacity (and respiration load, if the case) define the thermal loads during operation. Good practice in maintenance and in operation (i.e. adjustment of temperature setpoints according to the specific type of goods, adherence to loading limits, regular maintenance of the insulated box and refrigeration unit, appropriate pre-cooling of the storage compartment prior to loading, and the use of air curtains or plastic strips to minimize air infiltration during door openings), as reported in Devin and Cavalier (2020), contributes to the reduction of the overall thermal load.

For mechanically refrigerated equipment, the actual energy demand for a given thermal load is determined by the efficiency of the refrigeration unit. Consequently, enhancing unit efficiency represents a key strategy to reduce overall energy consumption. Although coefficients of performance (COP) are typically not widely disclosed by manufacturers, literature indicates that COP values for transport refrigeration units are generally lower than those of stationary refrigeration systems (Tassou et al., 2009; Wu et al., 2013).

Fabris et al., 2022, showed that decoupling the refrigerating unit from the vehicle driveline and implementing variable-speed control of the compressor can yield a 26.0 % improvement in mission COP and a 39.3 % increase in the mission duty cycle. Similarly, Artuso et al. (2019) quantified the degradation of the overall mission COP as a function of duty cycle. The adoption of inverter-driven electrical compressors therefore presents an opportunity to increase the average COP. In addition, refrigerated vehicles present a real chance of implementing solar panels on top of the roof, to support the refrigerating unit thus reducing the net demand from the vehicle or the electrical grid. This solution is increasingly promoted in the market. Rossetti et al., 2022, reported that, under Athens climatic conditions, rooftop solar panels can supply between 65 % and 112 % of the annual energy demand of the refrigeration unit, depending on the exposure time to direct solar radiation. Maiorino et al. (2024) demonstrated that the presence of PV panels and a dedicated battery pack can lead to a reduction in fuel consumption to power the refrigeration unit of >88 % in summer months, with an overall 33–47 % reduction in total carbon equivalent emissions.

It is noteworthy considering that reducing the energy demand from either the vehicle engine or the electrical grid directly contributes to lowering CO_2 emissions. However, given that the large majority of the commercial fleet is powered by an internal combustion engine (according to ACEA, 2025, only the 1.1 % of the LCV and 0.1 % of MHCV are electric vehicle in the EU area), the quantification of other pollutants, such as CO, THC, NO_x and PM should be addressed given their significant contribution to localized air pollution. Fabris et al., 2022, showed that, for their analysed case, the integration of a battery system to store energy generated by the engine, enabling a Start&Stop management operation, resulted in an 11.1 % reduction in fuel consumption. Moreover, this configuration achieved reductions of 24.0 %, 1.1 %, 25.0 %, and 16.6 % in CO, NO_x, THC, and PM emissions, respectively, when compared with a conventional configuration employing a direct belt connection between the engine shaft and the refrigeration unit.

Finally, the reduction of greenhouse gas emissions can also be pursued through interventions on refrigerant selection. Conventional synthetic refrigerants are associated either with high global warming potential (GWP) or with environmental concerns related to the production and persistence of PFAS substances. Consequently, international agreements and national regulations are progressively imposing stricter restrictions or bans on their use, with an overarching HFC phase-down perspective (Regulation (EU), 2024; The Kigali Amendment, 2016). Natural refrigerants then represent the long term environmentally sustainable solution as well as a reliable investment, considering that refrigerated vehicles have typically a 10–15 year long life (Tassou et al., 2009), with an average age of LCVs in use in Europe equal to 12.7 years (ACEA, 2025). In recent years, a relevant number of scientific and technical publications has demonstrated that natural refrigerants, particularly hydrocarbons (HCs) and carbon dioxide (CO_2), can outperform synthetic refrigerants in transport refrigeration applications. Colbourne et al. (2017), demonstrate that the COP of the R290 system is 15–25 % higher in MT applications (0 °C) and 10–30 % higher in LT applications (−20 °C) compared to the corresponding R404A system and that the R290 system resulted in -16 % diesel consumption and -66 % emissions. Fabris et al. (2024a), developed a dynamic numerical model to compare the energy performance of a R134a and a CO_2 unit in South European climatic conditions, showing that the seasonal COP of the CO2 unit is 27 % better than the R134a one, despite being characterized by a higher weight. The employment of light-weight components, such as a compact scroll compressor instead of a semi-hermetic piston compressor derived from stationary applications, and optimized control can then further reduce the overall carbon footprint of the CO_2 unit by approximately 18–35 %, depending on climatic conditions (Fabris et al., 2025).

A few research studies available in scientific literature have focused on the estimation of the environmental impact of transport refrigeration systems in terms of equivalent carbon emissions using a life-cycle assessment approach (Fabris et al., 2024a,; 2025; Li, 2017,; 2022a; Wu et al., 2013; Wu et al., 2022b). These research studies present different numerical models to assess the carbon footprint of a single Transport Refrigeration Unit (TRU) during their production, transport, use, repair and recycling phases. These studies highlighted that the main factors which influence the carbon footprint of TRUs over their life cycle consist of the indirect emissions linked to primary energy consumption during the cooling units operation, of the additional primary energy consumption required to transport the weight of the unit, and of the direct emissions due to refrigerant leakage, where the GWP of the refrigerant itself plays a crucial role. Possible improvements reported by such studies include some of the ones already mentioned above, such as the employment of natural refrigerants instead of synthetic working fluids (Colbourne et al., 2017; Fabris et al., 2024a,; 2025) and the integration of photovoltaic panels (Maiorino et al., 2024; Rossetti et al., 2022), but also other strategies such as the use of shore power for pre-cooling (Lin et al., 2025), the integration of thermal energy storage (Wu et al., 2022a), the use of cryogenic technology (Mohan and Amin, 2025), the reduction of the unit weight through selection of compact, light and efficient components (Fabris et al., 2025), and the integration of smart information technologies, such as Internet of Things, artificial intelligence and blockchain, in cold chain logistics (Han et al., 2026). However, all the above-mentioned studies focus on the performance and emissions of single TRUs; on the other hand, the scope of this research work is to provide an estimation of the carbon footprint of TRUs at continental fleet level, considering different climatic conditions, truck sizes, types of delivery mission and prime energy sources.

Therefore, in this paper, a methodology to estimate the emissions of

current EU[1] road transport refrigeration fleet is proposed and used to evaluate the reference emissions and compare them to an improved scenario, where the above anticipated strategies are implemented. The improvements adopted are representative of high TRL technologies which are rather available or close to becoming ready for the market, and which consequently have real chances of fast implementation.

2. Estimation of the EU refrigerated commercial fleet

The following paragraphs presents the available data and the procedure to estimate the overall refrigerated commercial fleet in the EU. In this manuscript, the following classification is adopted:

- Light Commercial Vehicle (LCV) up to 3.5 t;
- Medium, Heavy Commercial Vehicles (MHCV) over 3.5 t;
- Semitrailers (S).

A lowercase r prefixed to acronyms will be used to specify the refrigerated vehicles. Fleet estimations will be used in the results section to estimate the energy and carbon footprint of Transport Refrigeration Units (TRU) of each country. Complete results are presented in Table A1 inAppendix A.

2.1. Reference countries

The actual number of refrigerated trucks in service in Europe is not directly available in the open literature. Data collection was successful for Denmark, France, Germany, Italy, Poland, Spain and United Kingdom, although from different sources. From the available data, extrapolation has been carried out to obtain an estimate of the EU fleet.

The data for Denmark were kindly provided by the Danish ATP control authority (ATP-materielkontrollen). The France fleet estimation is based on the data published by Tecnea-Cemafroid in 2021 (Cavalier, 2021). Italian fleet data have been published by OITAF, the interdisciplinary observatory on food and drugs transport (Osservatorio Interdisciplinare Trasporto Alimenti e Farmaci (OITAF-1), 2022). Estimation of the Spanish refrigerated fleet has been provided by the "Ministerio de Industria y Turismo", based on information dispatched by the "Dirección General de Tráfico". German, Polish and UK's data were based on market survey and trade associations confidential communications.

Poland and UK data contained the number of refrigerated commercial vehicle registered annually. Given the units registered every year, the average age of the whole commercial fleet by vehicle type (ACEA, 2025) has been used to evaluate the total number of refrigerated vehicles in use in these countries, multiplying the registered vehicle by the average age.

2.2. Estimation of the remaining EU refrigerated fleet

The reference countries mentioned in the previous section represent a significant portion of the EU commercial fleet. In fact, according to the data published in ACEA, 2025, France, Germany, Italy, Poland, Spain, and UK are the countries in the EU area that own the largest commercial fleet in both the LCV (72 %, Fig.1a) and MHCV segments (66 %, Fig. 1b). These data support the assumption that the reference countries are a representative and relevant base to estimate the overall EU commercial refrigerated fleet.

The initial step in the estimation procedure is to associate the nominal load capacity (m) of vehicles to each different types of vehicles: 1.5 t was assumed for each rLCV (m_{LCV}) and 25 t as average load capacity for rMHCV+S (m_{MHCV+S}). The nominal load capacity is then multiplied by the estimated vehicles to obtain the national nominal refrigerated fleet capacity:

$$M_{r\ tot} = m_{LCV}\ N_{rLCV} + m_{MHCV+S}\ N_{rMHCV+S} \tag{1}$$

Eurostat provides figures about food products transported in each country of the EU by year (ϕ) in million tons-kilometre ("Food products, beverages and tobacco [GT04]" in Million tons-kilometre; EUROSTAT, 2024). As visible in Fig. 2, the countries with the largest fleet capacity (M_{tot}) (France, Germany, Italy, Poland, Spain, and UK) are confirmed to be those with the highest food product transport activities (ϕ). Denmark data set, on the other hand, confirms the presence of a proportionality between these two data sets.

The orange dots in Fig. 2 represent the estimated nominal refrigerated fleet capacity (in Mt) for the remaining Countries computed assuming a linear correlation best fitted on the available points:

$$M_{r\ tot} = \alpha\ \phi \tag{2}$$

where α is the proportionality coefficient obtained using the available refrigerated fleet data:

$$\alpha = \frac{\sum M^i_{r\ tot}}{\sum \phi^i}\ for\ i = DE,\ DK,\ ES,\ FR,\ IT,\ PL,\ UK \tag{3}$$

Accordingly, the average ratio of M_{rLCV} and $M_{r\ tot}$ has been computed as follows:

$$\beta = \frac{\sum M^i_{rLCV}}{\sum M^i_{r\ tot}}\ for\ i = DE,\ DK,\ ES,\ FR,\ IT,\ PL,\ UK \tag{4}$$

This ratio and the nominal loading capacity were then used to estimate the number of refrigerated vehicles for the European countries not included in the reference country list:

$$N_{rLCV} = \frac{\beta\ M_{r\ tot}}{m_{LCV}} \tag{5}$$

$$N_{rMCH+S} = \frac{(1-\beta)\ M_{r\ tot}}{m_{MHCV+S}} \tag{6}$$

2.3. Fleet composition

To characterize the fleet use in terms of temperature set, the share of refrigerated vehicles transporting chilled and frozen food has to be defined. In absence of reliable sources on this, an indirect approach has been adopted using the value of sold production, import, and export within the EU in 2019 (EUROSTAT, 2019). Fig. 3 shows the percentage of frozen and chilled food products for the reference countries as well as for the trend line considering EU+UK. According to this dataset, the chilled products represent 74 % of refrigerated food products in the whole EU area, while the remaining 26 % is represented by frozen food. These ratios have then been assumed to define the ratio between Medium Temperature (MT) and Low Temperature (LT) trucks for both LCV and MHCV+S.

3. Numerical model for emission evaluation

The emissions estimation for each country follows the steps presented in the flow chart in Fig. 4. The input data are first uploaded; they include the ambient conditions, the refrigeration box reference dimensions, the refrigeration unit performances and characteristics, and the energy source used to power the refrigeration unit. Then, the thermal load and thus the unit energy consumption are computed, according to the mission profile associated with the truck under consideration. Finally, using the prime engine and energy product data, energy

[1] The complete list of country considered in this paper include the EU-28 members, plus Norway and Switzerland: Austria; Belgium; Bulgaria; Croatia; Czechia; Denmark; Estonia; Finland; France; Germany; Greece; Hungary; Ireland; Italy; Latvia; Lithuania; Luxembourg; Netherlands; Norway; Poland; Portugal; Romania; Slovakia; Slovenia; Spain; Sweden; Switzerland; United Kingdom.

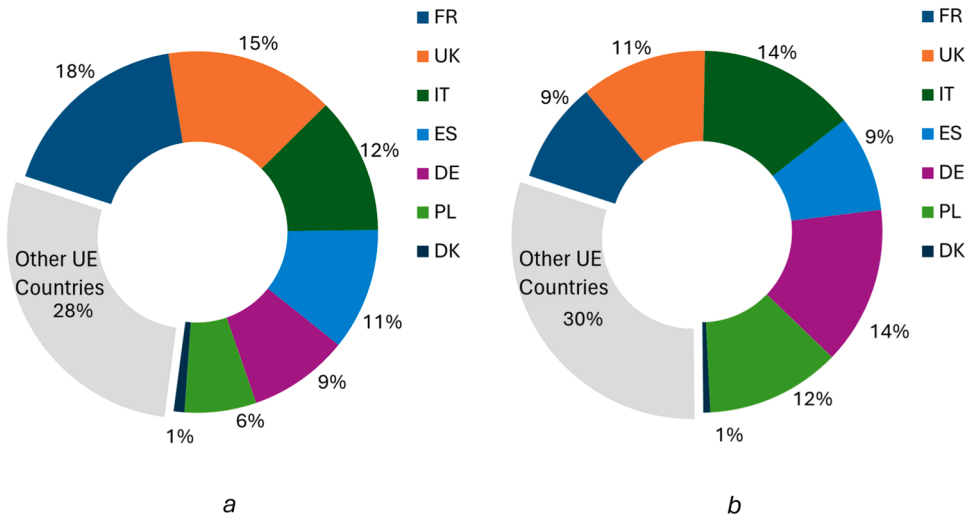

Fig. 1. Commercial European Fleet (including non-refrigerated trasnport); a) LCV; b) MHCV+*S* (ACEA, 2025, 2023 dataset).

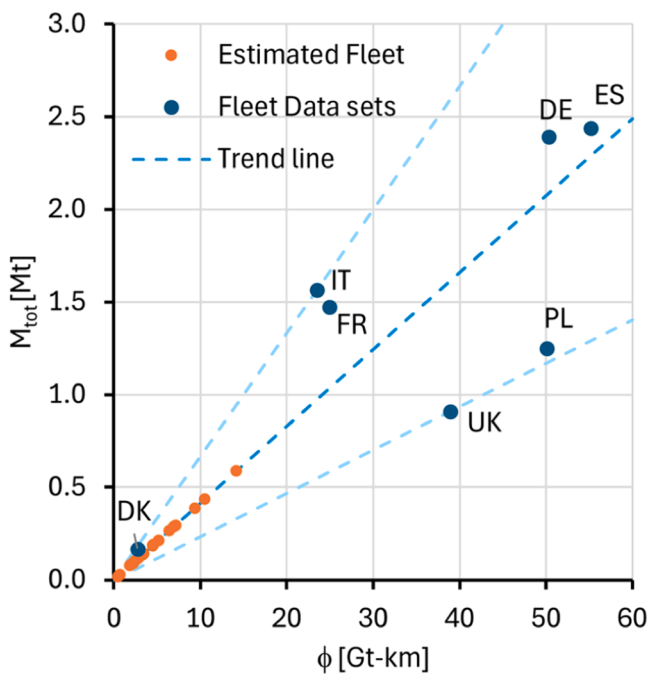

Fig. 2. Food products (GT04) compared to nominal refrigerated fleet capacity.

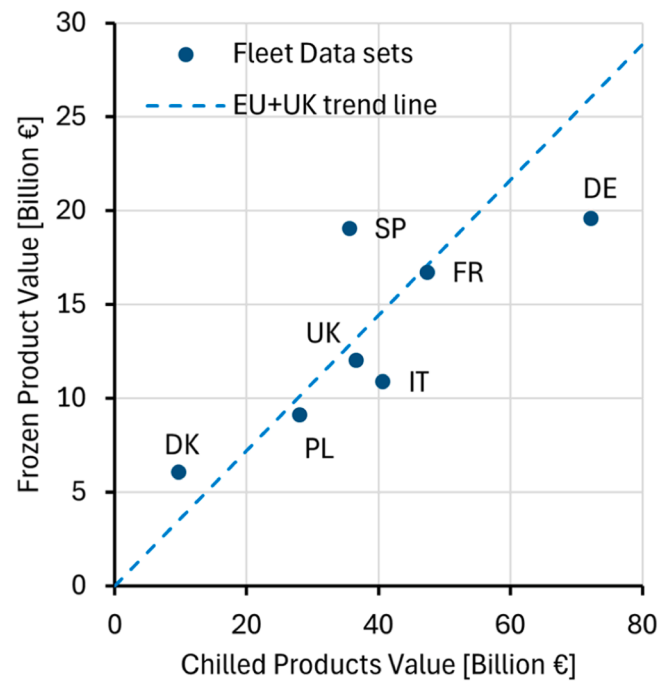

Fig. 3. Total value (production + import + export) of chilled and frozen products in the reference country and in EU+UK.

consumption is converted into pollutant and GHG emissions. Fugitive emissions from the refrigeration unit are also accounted at this step.

The mission profile is repeated for each day of the year, leading to the estimation of the yearly emissions for a single truck. The procedure is repeated for both vehicle classes (rLCV and rMHCV+*S*) and temperature classes (MT and LT). Values are multiplied by the corresponding number of trucks estimated in the country to obtain the year total for each considered country. This is then repeated for all EU+UK countries.

Each of these steps is presented in detail in the following sub sections.

3.1. Insulated box geometries

The average dimensions of the insulated box are assumed to be equal to 2.2 m x 2.1 m x 3.2 m (external surface area equal to 36.8 m^2) for rLCV and to 2.2 m x 2.5 m x 9.0 m (external surface area equal to 95.6 m^2) for the category rMHCV+*S*. They all have a double rear door covering the entire end wall of the box.

3.2. Climatic conditions

Epw files are used to describe the ambient conditions (Energy Plus Weather Data). Datasets include the hourly ambient temperature, air relative humidity, and solar radiation data for the reference year in each considered location. Generally, the capital city is considered as representative of each country, unless data are not available. In this case, data available for the closest city is taken.

3.3. Missions profiles

For the evaluation of the annual energy consumption of refrigeration units, it is assumed that each truck works 8 h/day, 5 days/week for 50 weeks (Li, 2017) all year long.

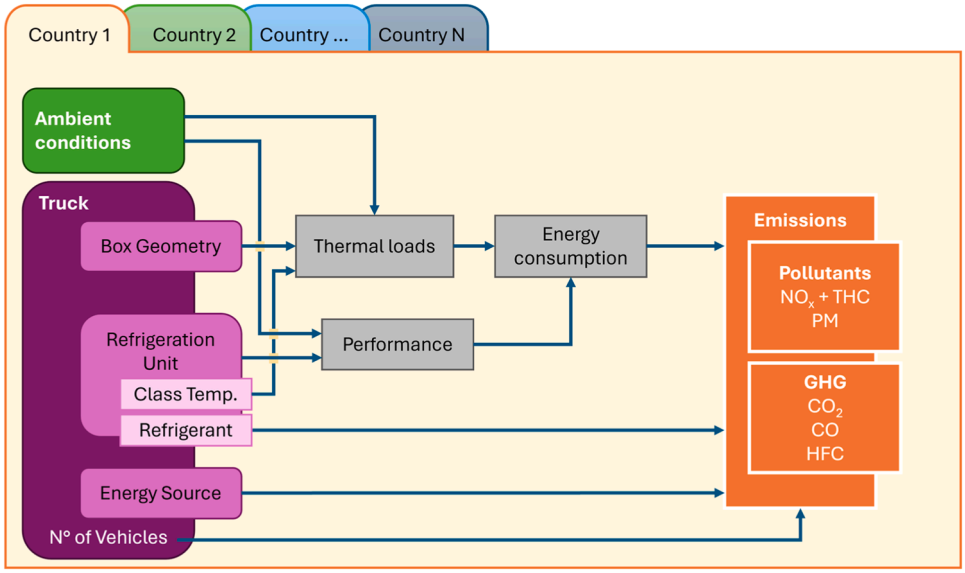

Fig. 4. Flow chart of the numerical model used to assess GHG emissions and pollutants.

Different reference missions are defined for rLCV and rMHCV+*S*. rLCVs are assumed to be mainly used for multi drop, short distance delivery missions, between 08:00 and 20:00 with 12 doors opening. Considering that the maximum working period consists of 8 h, missions start is assumed to be evenly distributed between 8:00 and 12:00 ending, accordingly, between 16:00 and 20:00.

Larger trucks are expected to be mainly employed for regional, national, and international transport with a uniform distribution of activity during the day (missions start between 1:00 and 24:00) and 4 doors opening. The opening time is 1 min for chilled freight and 30 s for frozen freight.

3.4. Refrigeration units

Two functions describe the Coefficient of Performance (COP) and maximum cooling power Q_{max} for the nominal set-point temperature T_{set} The refrigeration unit energy consumption is then computed for each hour as a function of the operating condition as:

$$P^h = \frac{Q^h_{demand}}{\varepsilon(z^h)\ COP(T^h_{amb})} \tag{7}$$

where Q^h_{demand} is the cooling demand, ε is the COP degradation factor caused by the duty cycle z:

$$z = min\left(\frac{Q_{demand}}{Q_{max}(T_{amb})}, 1\right) \tag{8}$$

The degradation factor ε formulation is adopted in accordance with UNI-EN-14,825:

$$\varepsilon(z) = 1 - C_d(1 - z) \tag{9}$$

with $C_d = 0.25$. This term was introduced to account for the degradation of performance due to cycling in part load conditions of ON—OFF units.

3.5. Thermal loads and refrigeration unit numerical model

The refrigerated box of each vehicle is assumed to be maintained at set-point temperature (T_{set}) during the whole mission: 0 °C for MT and -20 °C for LT. Ambient data has been used to quantify the main thermal loads: the pulldown of the insulated body, heat flux through the box walls and the infiltrations of ambient air.

The convection/conduction heat through the box walls was computed using the global heat transfer coefficient of the box *K*, the box average surface *S* and the temperature difference between the ambient temperature T_{amb} and the internal temperature T_{set}.

Imperfect sealings and door openings cause external air infiltration, resulting in infiltration heat load. The external air leaking inside the box due to the infiltration on the sealings w_{lk} is assumed to be 0.02 times the box volume per hour according to Li et al. (2010). The air mass entering the truck during the doors openings w_{op} derives from a simplified polynomial approximation of the experimental results reported by Lafaye de Micheaux et al. (2015). Sensible and latent heat load are then computed given the humid air enthalpy at ambient (h^h_{amb}) and set point (h_{set}) conditions.

For each hour during the mission the thermal loads are computed as follows:

$$Q^h_{loads} = kS\left(T^h_{amb} - T_{set}\right) + \left(w^h_{op} + w^h_{lk}\right)\left(h^h_{amb} - h_{set}\right) \tag{10}$$

The cooling demand is obtained limiting the cooling loads to the maximum cooling capacity of the unit:

$$Q^h_{demand} = min\left(Q^h_{loads};\ Q_{max}\left(T^h_{amb}\right)\right) \tag{11}$$

The pulldown of the refrigerated box is assumed to occur before the mission begins, and before any freight is loaded in the box. The pulldown energy of the refrigerated box U_{box} is estimated as the energy required to cool down the whole refrigerated box from the daily average temperature to the equilibrium working temperature. Assuming a linear temperature profile between the external and the internal wall, this is estimated as the average between the external temperature and the set-point temperature. The thermal capacity of the box is assumed to be linear with the box surface. The proportionality constant is defined in agreement with the experimental data reported by Artuso et al. (2019b).

The pulldown energy of the humid air inside the box U_{air} is computed assuming the mass of air contained in the truck and its enthalpy change from ambient condition to set-point temperature.

The pulldown energy *U* measures the energy needed to cool the box down to its operative temperature and, as pulldown is concluded before the mission starts, it is set to zero at the start of the mission and set to ($U_{box} + U_{air}$) at the beginning of the pulldown. The duration of the pulldown is obtained computing the hourly pulldown energy to be provided (U^h) backwards in the hours preceding the mission, according to:

$$U^h = min\left(U_{box} + U_{air}; U^{h+1} + 3600\ \left(Q^h_{max}\left(T^h_{amb}\right) - Q^h_{loads}\right)\right)$$

$$with\ h = 0,\ -1 \ldots\ until\ U^h = U_{box} + U_{air} \tag{12}$$

The pulldown load for the hours in which the pulldown occurs is then calculated as:

$$Q^h_{demand} = \frac{U^h - U^{h+1}}{3600} \tag{13}$$

It is worth noting that thermal loads Q^h_{loads} (namely thermal and ambient air infiltrations, see Eq. (10)) are accounted also during the pulldown.

3.6. GHG and other pollutants

The environmental impact of the unit depends on the physical energy source. For this reason, different conversion factors are used to assess CO_2 emissions as well as the main pollutants, according to the system layout, as reported in Table 1.

Diesel engines are assumed to be coupled to the compressor by means of direct mechanical transmission (belt driven) or using a generator mechanically connected to the engine to provide electrical power to a hermetic or semi-hermetic compressor.

Grid Electricity is assumed to be stored in dedicated battery packs when the vehicle is off duty to be then delivered to a hermetic or semi hermetic compressor when the cooling unit is turned on. Transmissions efficiencies and battery charge-discharge efficiency are neglected.

To evaluate the overall environmental impact of the cooling unit, the direct emissions associated with the refrigerant leakage are considered. Annual leakage of 15 % of the nominal charge was assumed on this work, as a reasonable value inside the estimation reported in literature by different authors (10–37 % Tassou et al., 2009, 10–25 % Li, 2017, 5–25 % Wu et al., 2022b). The Global Warming Potential (GWP) are estimated according to IPCC (2021).

4. The baseline and the sustainable scenarios

The present work compares a baseline scenario, representing solutions commonly adopted in the actual European commercial refrigerated fleet, with more sustainable scenarios where the cumulative adoption of environmental-friendly solutions is considered.

4.1. Baseline

The baseline scenario considers synthetic refrigerant cooling units. Unit is controlled by an ON/OFF thermostat with hysteresis. Nominal cooling powers at 30 °C ambient temperature have been selected according to commercial catalogues (Daikin, 2024), and are reported in Table 2.

The COP of the baseline units is modelled according to the data published by Colbourne et al. (2017). While the abovementioned paper refers to a R404A, it is assumed these values are representative of retrofitted units using R452A.

As reported by Cavalier et al. (2023) the great majority of boxes (>93 %) are produced so to be compliant with ATP regulation (ATP, 2024) for LT transport, which prescribes the global heat transfer coefficient to be lower than 0.40 W m^{-2} K^{-1}. A global heat transfer coefficient 5 % lower than the one prescribed is assumed for LT: K_{LT} = 0.38 W m^{-2} K^{-1}. This is coherent with the distribution of K values of prototype in Cavalier et al. (2023) where the most populated category (63 %) in the frequency distribution histogram is $0.35 < K < 0.40$.

The limit for MT equipment is set by ATP to 0.7 W m^{-2} K^{-1}. According to Cavalier et al. (2023) only 6 % boxes are specifically designed to meet this requirement. In most cases MT transport is performed using equipment designed for LT transport that is no longer compliant with this type of transport, as a result of aging. Capo et al. (2019) reported that the aging (first 12 years) increases the K values by 35 %, 45 % and 44 % for LCV and MHCV and Semitrailer respectively. This study is representative of average trucks lifetime in EU, as reported by ACEA (2025), that is 12.7 years for LCV and 14.1 years for MHCV. The average K value for MT boxes is then computed assuming the K value at the year of production is coherent with K_{LT} increased by the above-mentioned aging factors. The global heat transfer coefficients used for different trucks and temperatures are summarized in Table 3.

Diesel driven engines are used as prime movers for all the baseline refrigeration units. A dedicated engine (NRMM Stage V) is assumed to be used for larger applications (MHCV+*S*) while in case of LCV the compressor is assumed to be run deriving mechanical power from the vehicle engine (Euro VI) by means of belt-driven electric clutch.

Table 1
Conversion factor for different energy products and prime mover. Sources: a) Grigoratos et al., 2019,b) Desouza et al., 2020; d) EEA, 2022 c) ICCT, 2016 e) ISPRA, 2020 ‡) emissions for the electrical grid are expressed in terms of volatile organic compound instead of total hydrocarbons.

Energy product	Engine/motor type	Emissions [g/kWh]			
		CO_2	CO	NO_x+THC	PM
Diesel	Vehicle engine (EURO VI)	651 [a]	0.88 [a]	0.45 [a]	0.01 [a]
	Dedicated engine (NRMM Stage V)	575 [b]	6.60 [c]	7.50 [c]	0.40 [c]
Grid Electricity	Electrical motor	238 [d]	0.09 [e]	0.30 [e ‡]	0.00 [e]

Table 2
Rated cooling power of the cooling units for different vehicles at 30 °C ambient temperature.

Application (Set point temperature)	LCV Q_{design} [W]	MHCV+*S* Q_{design} [W]
MT (0 °C)	4700	8600
LT (-20 °C)	2470	4700

4.2. Improved solutions

Different improvements toward sustainability are considered by acting on different aspects affecting the unit carbon footprint. Each is presented in the following paragraphs.

4.2.1. Improved insulation - Imp(K)

The global heat transfer coefficient of a refrigerated truck can reach values as low as 0.27 W m^{-2} K^{-1} (Capo et al., 2020). For this paper, an average global heat transfer coefficient of 0.32 W m^{-2} K^{-1} will be considered an achievable target for the EU fleet (-20 % in respect to ATP limit). MT will be assumed to be aged starting from this improved value, and aging factors are halved in respect of the baseline scenario, assuming improved maintenance, and a reduction of the average age of the fleet. Resulting global heat transfer coefficients in the improved insulation scenario, for different trucks and temperatures are summarized in Table 4.

Table 3
Global heat transfer coefficient in the actual EU fleet, per truck type and application.

Application (Set point temperature)	LCV K [W m^{-2} K^{-1}]	MHCV+*S* K [W m^{-2} K^{-1}]
MT (0 °C)	0.51	0.55
LT (-20 °C)	0.38	0.38

Table 4
Global heat transfer coefficient in the improved insulation scenario, per truck type and application.

Application (Set point temperature)	LCV K [W m^{-2} K^{-1}]	MHCV+S K [W m^{-2} K^{-1}]
MT (0 °C)	0.38	0.39
LT (-20 °C)	0.32	0.32

4.2.2. Natural working fluids - NWF

R290 units is considered as sustainable replacements of the baseline units, with performance taken from Colbourne et al. (2017). R744 applications have been proposed in literature for LCV (Fabris et al., 2024a, 2024b), but in absence of consolidated experimental characterization they are not considered here.

4.2.3. Improved control at partial loads - Imp(CNT)

To improve the unit performance at partial loads, the use of a variable-speed compressor is considered. This improvement assumes that the unit can modulate its cooling power down to 25 % of the rated value before entering the ON/OFF control. As this control implies the use of an inverter driven compressor, coupling between the refrigeration unit and its energy source is performed in this case using an electrical generator and a small battery to stabilize the system and dump small unbalances between electrical energy production and consumption. The benefit of this solution has been demonstrated in (Fabris et al., 2022).

In this scenario Eq. (5) is rewritten to account for this feature, limiting the ON/OFF regulation to the case of $Q_{demand} < 0.25\ Q_{max}$, modifying the duty cycle definition accordingly:

$$z = min\left(\frac{Q_{demand}}{0.25\ Q_{max}(T_{amb})}, 1\right) \tag{14}$$

4.2.4. Improved prime engine - Imp(ENG)

Electrification can play an important role in the sustainability of small refrigeration units applied to rLCV. In these applications the energy consumed by the unit during a daily mission can be reasonably stored in high-capacity batteries. As the battery is charged using the grid electricity when the vehicle is out-of-duty, average EU grid conversion factor is used to assess the unit emissions in rLCV.

While the same approach is, in principle, feasible for larger applications (rMHCV+S), achieving complete autonomy on a long-haul journey by means of high-capacity battery can be challenging in terms of investment cost and weight. Thus, for MHCV+S the improved solution assumes the use of EURO VI engine instead of a NMRR engine, to reduce the pollutant emissions.

4.2.5. Integrated photovoltaic panels

Photovoltaic panels (PV) can be integrated into the insulated box roof to provide a quota of renewable energy to the refrigeration system, reducing the net energy consumption requested by the unit. PV efficiency η_{PV} is assumed to be equal to 20 %.

Global Horizontal Radiation (GHR) from the ambient data set is used to compute the collected power. The daily variation of the solar radiation (day – night) during the year as well as whether conditions (clouds cover) are inherently considered, as the weather data files already include these factors.

Nevertheless, urban canyon and other shading effects can reduce the quantity of the direct solar radiation collected by the solar panels (Rossetti et al., 2022). A decreasing coefficient $k_{GHR} < 1$ is included in the equation to account for this. Moreover, PV involves an effective area lower than roof area A_{roof}. This is accounted by using the factor $k_a = A_{PV}/A_{roof} \leq 1$. The electrical power produced by the PV is then assessed as follows:

$$P^h_{PV} = \left(k_{GHR}\ GHR^h\right)\ \left(k_A\ A\right)\ \eta_{PV} \tag{15}$$

The electrical system is assumed to have negligible storage capacity. The balance between consumption and production is performed hourly. In case the PV at a given hour produces more energy than the one consumed by the unit, the excess energy is not stored. This hypothesis underestimates the energy saving that can be obtained implementing PVs, in order to favour a simpler and lighter layout of the system.

As results of these assumptions Eq. (8), assessing the unit power drawn from the primary power source, is in this case substituted by the following:

$$P^h = max\left(\frac{Q^h_{demand}}{\varepsilon(z^h)\ COP(T^h_{amb})} - P^h_{PV}, 0\right) \tag{16}$$

5. Results

5.1. Baseline situation

The baseline scenario is presented in Fig. 5. The emission of each country (Fig. 5a) highlights the impact of the most numerous fleets, with the 5 largest fleet clearly above the average of the remaining countries. Overall emissions are also influenced by the latitude, with southern countries characterized by higher emissions: while the German and the Spanish fleet are similar in capacity, the latter has a significant higher GHG impact due to the higher average ambient temperature.

To quantify the relative impact of the climatic conditions and fleet size at country level, Fig. 5b reports the emission of the average truck in each country, while Fig. 5c reports the fleet size in terms of nominal capacity (according to Eq. (1)). The ratio in terms of specific emissions between the highest (Greece, 2.85 tCO_{2e}/truck/year) and lowest (Finland, 1.91 tCO_{2e}/truck/year) value is limited to 1.5, while ratio between the largest fleet (Spain, 2.44 Mt) and the smallest (Estonia: 0.02 Mt) is 122, demonstrating the fleet size is, by far, the most impacting factor.

The impact of the improvements is presented in the next section at the EU level. In these comparisons, the fleet size for each country will be considered constant to maintain the focus on the technical aspects of refrigerated transport. Nevertheless, it is worth noting that, as suggested by Fig. 4c, intervening in the fleet size by implementing new and more efficient logistics is a direct action to improve the cold chain sustainability.

5.2. System improvements: insulation, NWF, control and prime engine

The annual primary energy consumption of the European road refrigerated transport fleet for the baseline and different improvements scenarios as defined in 4.2.1- 4.2.3 is reported in Fig. 6, classified by temperature level (MT and LT) and vehicle class (rLCV and rMHCV+S).

The implementation of the improved insulation scenario leads to an energy demand reduction of 10.1 % when the whole EU fleet is considered. The shift to NWF TRU can add 9.5 % to this saving, while the control of the compressor speed contributes with a saving of 8.1 %. The best-case scenario, in which all these modifications are implemented, leads to 27.7 % less energy consumption.

The annual CO_2 emissions linked to the operation of the cooling unit, according to the energy demand of the unit and the prime mover used to run it, are reported in Fig. 7a where the improved scenario for the primal engine is also considered (4.2.4).

According to the CO_2 emission factors in Table 1, the switch from a dedicated NRMM diesel engine to Euro VI engine as the prime mover in MHCV+S might seem counterproductive, as it leads to a moderate increase in the operational emissions. However, this switch is justified by the massive reduction in the emission factor of other pollutants, as discussed further below. Despite this increase, the reduction of primary energy associated with the NWF cooling units and the average reduction of CO_2 emissions factors due to electrification of the LCV units (see

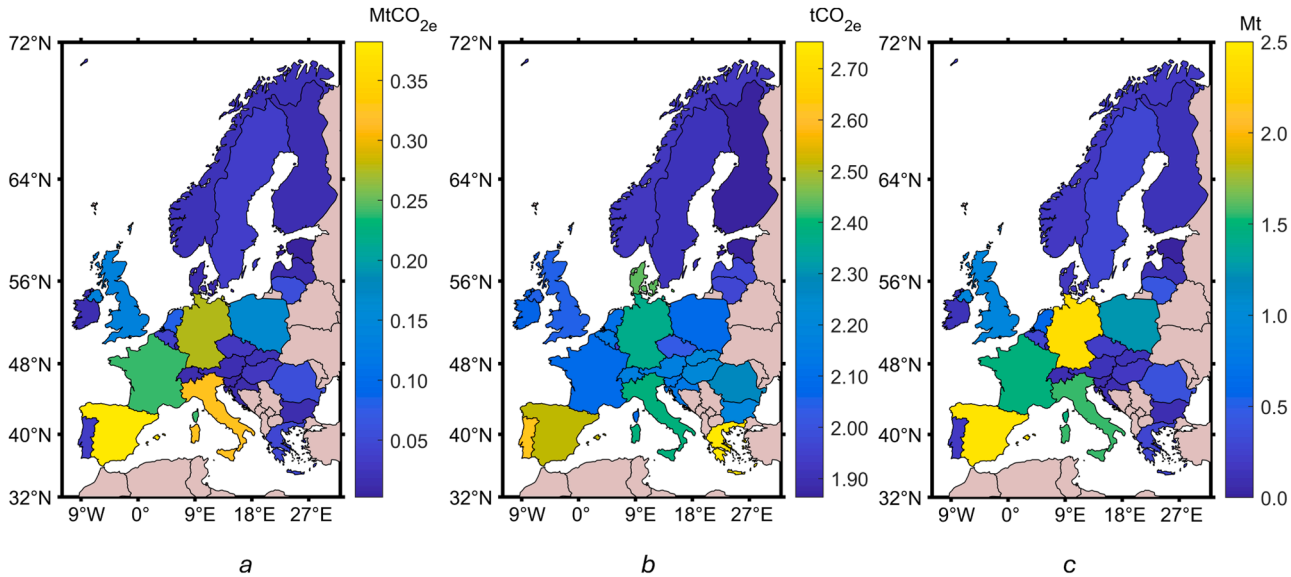

Fig. 5. The baseline EU emission due to refrigerated units in road transport: a) country yearly GHG emissions; b) yearly average emissions per single truck; c) fleet size (nominal fleet capacity).

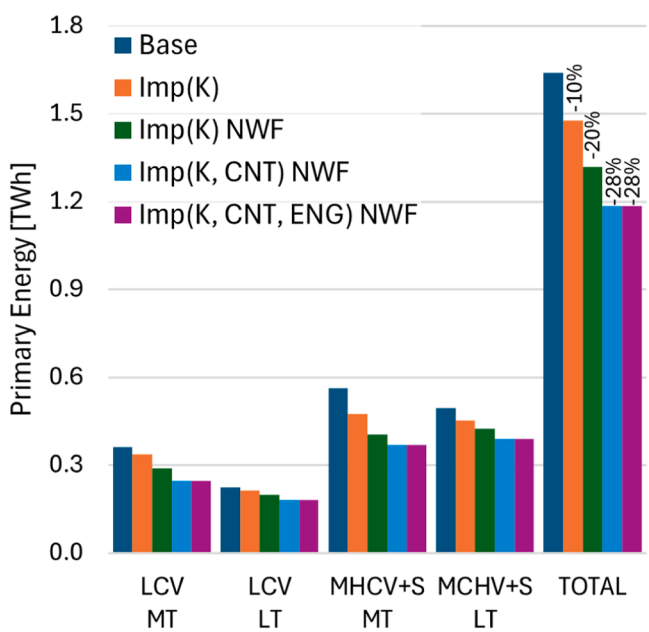

Fig. 6. - Annual primary energy consumption for the operation of European road refrigerated transport sector in different scenarios.

Table 2) lead to an overall 39.8 % reduction of annual CO_2 emissions in the best-case scenario (Fig. 7a).

The positive impact of NWF units is even more significant when the refrigerant leakages, converted in terms of CO_2 equivalent emissions, are taken into consideration (Fig. 7b). The energy saving and the negligible GHG potential of NWF lead to 57.9 % reduction thanks to the NWF unit implementation. An additional -10 % of emissions can be achieved by the implementation of a better control system and electrification of the light commercial fleet. Residual emissions in the best-case scenario are reduced by 72 % and are mainly caused by operational emissions of MHCV+*S* which are the only units running on fossil fuels.

As above mentioned, emissions of non-direct GHG pollutants must be taken into consideration, especially in densely populated areas, given their impact on human health. Therefore, the annual emissions of the main pollutants (carbon monoxide, CO, nitrogen oxides and total hydrocarbon, NO_x + THC, particulate matter, PM) are reported in Fig. 8. NRMM are characterized by very high pollutant emissions and, consequently, the greatest part of pollutant emissions reduction is linked to the switch to a better primary energy source for MHCV+*S*. On the other hand, the complete electrification of LCV units, despite being small in absolute terms when compared to MHCV+*S*, can reduce localized pollution in urban areas and comply with zero emission zones. Overall, in the best-case scenario, annual CO emissions can be reduced by 90.6 %, NOx and THC emissions by 94.3 % and PM emissions by 98.2 %.

5.3. System improvements: photovoltaic panels

The contribution of roof integrated PV panels, as defined in 4.2.5, is discussed hereafter. As both the different shares of direct solar radiation (k_{GHR}) and the ratios between the available area and the area covered by the PV panels (k_A) decrease proportionally the solar power production, results will be presented for the sake of simplicity considering these two elements grouped in one overall diminishing factor:

$$k_{PV} = k_{GHR}\, k_a \tag{17}$$

The sensitivity to this parameter is analysed for three cases: k_{PV} = 0.2, 0.5 and 0.9. The highest value (0.9) represents an optimistic scenario in which the solar panel fully covers the box roof, and the truck operates predominantly with an unobstructed view of the sky. Conversely, the lower value (0.2) may correspond to a smaller photovoltaic (PV) system and/or frequent obstructions to direct sunlight due to the nature of the truck's operation, for instance, extended delivery missions in an urban canyon environment.

The impact of PV on the total emissions is presented in Fig. 9, where in Fig. 9a solar panels are integrated to the baseline solutions while, in Fig. 9b, the PV panels are integrated into the best-case scenario discussed in Section 5.2.

The absolute values of overall energy savings for the entire EU fleet are higher when solar panels are integrated into baseline technologies rather than into the best-case scenario. Baseline technologies consume more energy than the best-case scenario, making it more likely that the energy generated by the solar panel will be fully utilized by the system. Additionally, the higher conversion factors of LCV refrigeration units in the baseline scenario amplify energy savings when converted into emissions reductions.

Despite the lower absolute savings, PV implementation remains

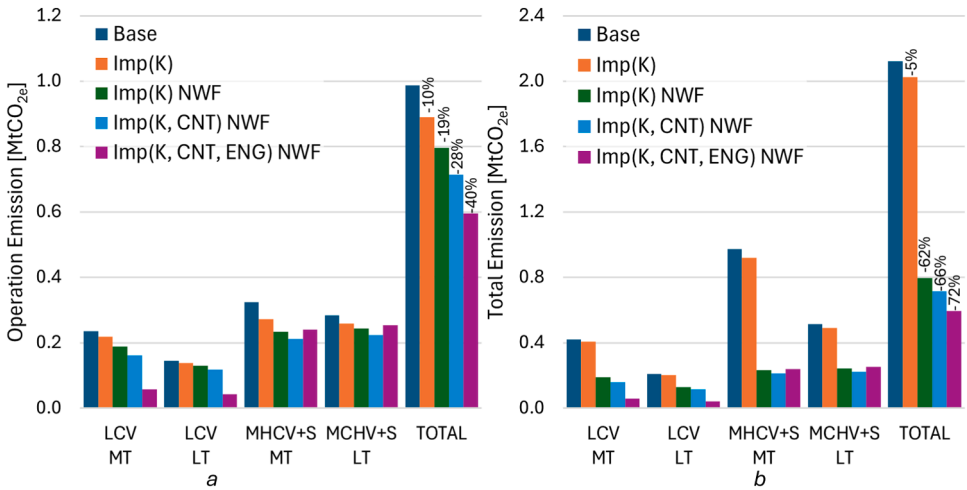

Fig. 7. GHG emissions of the European road refrigerated transport sector: a) operation emissions; b) total emissions (including operation and refrigerant leakage).

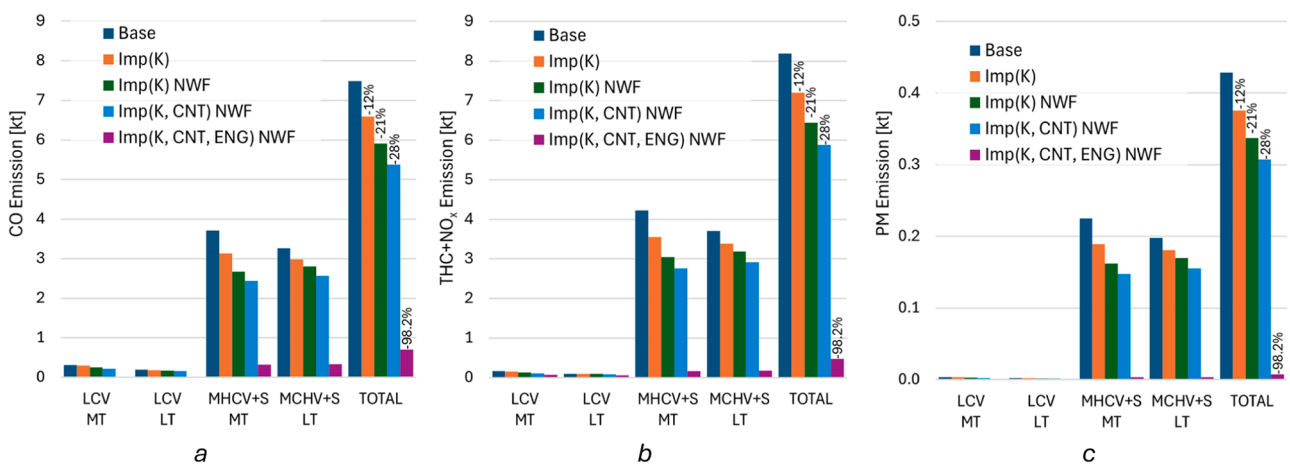

Fig. 8. Polluant emissions of the European road refrigerated transport sector: a) CO; b) THC+NO_x; c) PM.

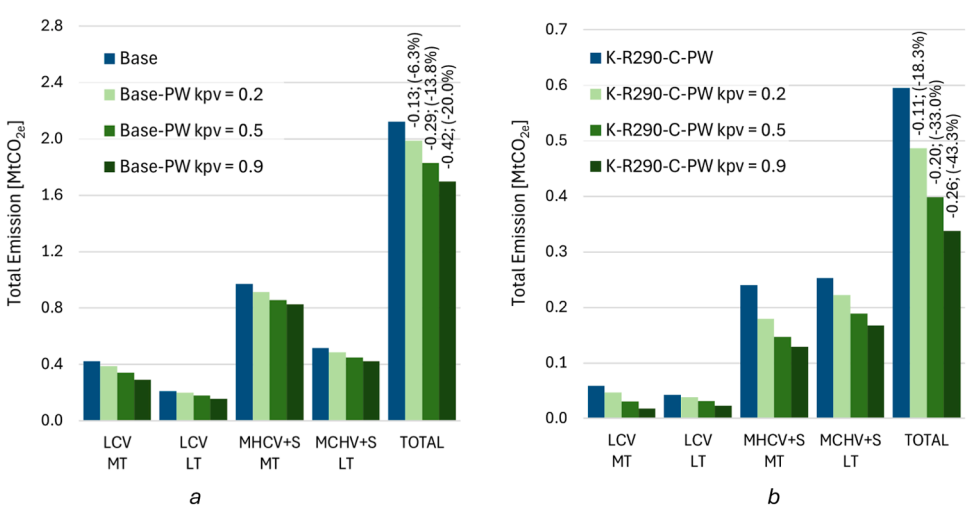

Fig. 9. Impact of PV panels on the total GHG emissions: a) using the baseline scenario as reference; b) usinge the best-case scenario as reference.

promising even in the best-case scenario. Emissions reductions in the MHCV+*S* group are consistent with those achieved in the baseline case and are relatively more impactful, reaching up to a 46 % decrease. In contrast, the absolute emissions savings for LCV units are minimal because, in the best-case scenario, these units rely on grid electricity, which has a lower emissions factor than diesel engines. Nevertheless, in this case, PV can enhance unit autonomy and reduce weight by allowing for a downsized battery pack.

It is important to note that these results are based on the limiting assumption that PV-generated energy must be consumed within the same hour at which it is produced; otherwise, it is wasted (Eq. (16)). Additionally, the MHCV+*S* fleet is assumed to operate with equal frequency during nighttime and daytime. Introducing a dedicated battery pack to enable non-synchronous energy production and consumption could significantly enhance the savings associated with PV implementation and reshape optimization strategies. While Rossetti et al. (2022) have explored some aspects of this, further investigation is required to gain a clearer understanding of PV's potential in refrigerated road transport. Conducting economic and life cycle analyses (LCA) would be essential to assess the sustainability of this solution and to define precise application boundaries, including truck size, PV system coverage, investment costs, payback period, operational scenarios (daily, nocturnal, or mixed), set-point temperatures, and possible integration with hybrid or electrified vehicles.

6. Conclusions

This manuscript estimates the environmental impact of the transport refrigeration sectors in EU and quantitatively assesses the possibilities to decrease the environmental impact by implementing incremental improvements to the refrigerated equipment.

The implementation of efficient solutions (such as better box insulation, natural refrigerant units, more sustainable energy sources and improved unit control) can lead to a reduction of the annual energy consumption of the European fleet up to 28 %, corresponding to a saving of CO2 equivalent emissions up to 72 %. Other pollutants can be reduced to >90 %. Complete transition to NWF by itself can reduce the emissions of 58 % thanks to the negligible GWP of NWF compared to synthetic refrigerants. Solar panels can significantly contribute to the energy balance of the truck and thus the overall annual emissions reduction also under the assumption of small capacity energy storage, both when applied to baseline or best-case technologies.

More detailed analysis, like LCA methodology, can highlight weak points and suggest proper actions: Fabris et al., 2024, suggested that refrigerating weight reduction by dedicated lightweight components application is a key factor to further decrease the total lifetime CO2 equivalent emissions of the analysed R744 unit, which was already 26 % less emitting than the baseline R134a unit.

Further research dealing with the technical aspects of the topic is then needed to include the following, to properly identify global optimality toward sustainability:

- the impact of the proposed improvements in terms of system weight, which impacts both on the loading capacity and on the energy consumption of the vehicle driveline;
- the inclusion of electrical accumulators to store the energy produced by PV panels to increase the share of renewable energy consumed by the refrigeration unit;
- LCA assessment of material and components.

Future work should also address the economic implications of these technological solutions, and the assessment should be based on the expected adoption rate of these solutions since unit costs are strongly influenced by production scale. Such adoption rates, however, depend on socio-economic policies supporting sustainability in the transport sector, as well as regulatory constraints such as the F-gas regulation and other legislation limiting the use of synthetic refrigerants.

Nomenclature

Symbols		
A	insulated box roof area	m^2
C_d	performance degradation factor	-
h	enthalpy	kJ kg^{-1}
k	partializaton factor	-
K	global heat transfer coefficient of the box	W m^{-2} K^{-1}
M_{tot}	nominal fleet capacity	t
m	nominal load capacity	t
N	number of vehicles	-
P	power consumption	W
Q	thermal flux	W
S	insulated box average surface	m^2
T	temperature	K
U	thermal capacity	kJ
w	air infiltration mass flow	kg s^{-1}
z	duty cycle	-
α	proportionality coefficient between the food products transport volume and the nominal refrigerated fleet capacity	m^{-1}
β	proportionality coefficient between light refrigerated commercial vehicles nominal capacity and the nominal refrigerated fleet capacity	-
Φ	food products transport volume	Mt km
η	conversion efficiency	-
Superscripts		
h	referring to the h hour in the numerical model	
Subscripts		
air	air	
amb	ambient temperature	
box	refrigerated box	
demand	referring to the thermal loads	
LCV	Light commercial vehicles	
lk	leakage due to sealing infiltration	
loads	referring to thermal loads	
max	cooling unit maximum capacity	
MHCV	Medium and Heavy commercial vehicles	
MHCV+*S*	Medium and Heavy commercial vehicles plus semitrailers	
op	opening	
PV	photovoltaic panels	
r	refrigerated	
S	Semitrailers	
set	set point	
Acronyms and abbreviations		
CNT	control	
COP	Coefficient of Performance	
ENG	Prime engine	
GHG	Greenhouse Gasses	
GHR	Global horizonal radiation	
GWP	Global Warming Potential	
IMP	improved	
LCV	Light commercial vehicles	
LT	Low temperature	
MHCV	Medium and Heavy commercial vehicles	
MT	Medium temperature	
NWF	Natural Working Fluids	
S	Semitrailers	
TRU	Transport Refrigeration Unit	

CRediT authorship contribution statement

Antonio Rossetti: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Francesco Fabris:** Writing – review & editing, Writing – original draft. **Sergio Marinetti:** Writing – review & editing. **Silvia Minetto:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization.

Declaration of interest statement

The author is an Editorial Board Member/Editor-in-Chief/Associate Editor/Guest Editor for this journal and was not involved in the

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Appendix A

Table A1
Estimation of the refrigerate fleet in EU28+CH+NO; Reference countries' data (bold) are not interpolated but collected from different sources are reported into the paper.

Country		Nominal Refrigerated Fleet Capacity [Mt]	Number of LCV	Number of MHCV+*S*
AT	Austria	0.13	3582	4982
BE	Belgium	0.29	7926	11,023
BG	Bulgaria	0.11	2905	4040
HR	Croatia	0.09	2488	3461
CZ	Czechia	0.21	5877	8174
DK	**Denmark**	**0.17**	**70**	**6690**
EE	Estonia	0.02	538	748
FI	Finland	0.14	3904	5429
FR	**France**	**1.47**	**60,094**	**55,368**
DE	**Germany**	**2.39**	**22,709**	**94,276**
EL	Greece	0.27	7334	10,199
HU	Hungary	0.19	5257	7311
IE	Ireland	0.13	3592	4996
IT	**Italy**	**1.56**	**77,976**	**57,871**
LV	Latvia	0.12	3254	4525
LT	Lithuania	0.44	12,043	16,748
LU	Luxembourg	0.03	808	1123
NL	Netherlands	0.59	16,218	22,554
NO	Norway	0.19	5124	7127
PL	**Poland**	**1.25**	**30,505**	**48,166**
PT	Portugal	0.19	5163	7181
RO	Romania	0.39	10,693	14,871
SK	Slovakia	0.14	3794	5276
SI	Slovenia	0.08	2122	2952
ES	**Spain**	**2.44**	**58,000**	**94,000**
SE	Sweden	0.30	8144	11,326
CH	Switzerland	0.09	2570	3574
UK	**United Kingdom**	**0.91**	**31,691**	**34,485**
Total EU28+CH+NO		236.36	394,381	548,473

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